

The Spectral Transformation Lanczos Algorithm for the Symmetric-Definite Generalized
Eigenvalue Problem: A Comparative Analysis with Conditioning Insights

by

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ABSTRACT

This thesis investigates the application of the spectral transformation Lanczos algorithm (ST-Lanczos) to a generalized symmetric-definite eigenvalue problem involving real symmetric matrices A and B , with B being positive definite and possibly ill conditioned. The Lanczos algorithm is a well-known iterative algorithm for computing the eigenvalues of a symmetric matrix and it works well for finding the extreme points in the spectrum. By leveraging a shifted and inverted formulation of the problem, the ST-Lanczos algorithm relies on iterative projection to approximate extremal eigenvalues near a shift σ . While previous work has been done using direct methods, the goal of this thesis is to use an iterative approach, and analyze how the error bounds already proven for direct methods play out in an iterative context.

This study focuses primarily on benchmarking the ST-Lanczos method against established direct methods in the literature and addresses challenges in numerical stability, computational efficiency, and sensitivity of residuals to ill-conditioning.

INDEX WORDS: eigenvalues, eigenvectors, Lanczos algorithm, Ritz values, Krylov subspaces, spectral transformation, orthogonality

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DEDICATION

To my loving parents, Mr. and Mrs. Adebessin —This work is a tribute to the sacrifices you made, the lessons you taught me, and for giving up your dreams so that I could chase mine. For offering me your best, your love and the boundless faith you have in me.

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CHAPTER 1

INTRODUCTION

1.1 Background

The problem of computing eigenvalues and eigenvectors of matrices in numerical linear algebra is a well-studied one. The computation of eigenvalues and eigenvectors plays a central role in scientific computing with applications in structural analysis, quantum mechanics, data science and control theory. However, eigenvalue problems (standard and generalized) involving dense and sparse matrices present significant computational challenges, especially as the size of the matrices increases. These problems are fundamental in many scientific and engineering disciplines where the underlying mathematical models are often expressed in terms of eigenvalue equations. Historically, methods for solving eigenvalue problems date back to the early 20th century, with foundational contributions from David Hilbert, Erhard Schmidt, and John von Neumann, who laid the groundwork for understanding linear operators and their spectral properties [2].

With the advent of digital computing in the mid-20th century, numerical methods for eigenvalue problems began to flourish. Classical iterative methods, such as the power iteration and inverse iteration, were among the first to be employed due to their simplicity and effectiveness for small-scale problems. However, as computational requirements grew, particularly with the need to solve larger sparse systems, researchers turned to more sophisticated algorithms. The Lanczos method, described in [4], introduced by Cornelius Lanczos in 1950, represented a significant advancement for efficiently solving eigenvalue problems

for large symmetric matrices. The method exploits the sparsity of matrices and reduces the dimensionality of the problem by constructing a tridiagonal matrix whose eigenvalues approximate those of the original matrix.

An important class of eigenvalue problems, which is the main focus of this thesis, is the generalized eigenvalue problem (GEP). The GEP takes the form $A\mathbf{v} = \lambda B\mathbf{v}$ where A and B are square matrices, λ is a generalized eigenvalue, and $\mathbf{v} \neq \mathbf{0}$ is the corresponding generalized eigenvector. This class of problems arises naturally in a number of application areas, including structural dynamics, data analysis and has a long history in the research literature on numerical linear algebra.

1.2 Literature Review

Generalized eigenvalue problems involving symmetric and positive definite matrices are fundamental in numerical linear algebra with applications in structural dynamics, quantum mechanics, and control theory. Solving these kind of problems involve computing the eigenvalues λ and eigenvectors $\mathbf{v}$ that satisfies the equation. The choice of method depends on the properties of the matrix involved in the problem we are trying to solve (e.g, sparsity, symmetry) and computational constraints. In this section, we discuss some of the research that has been done on this topic.

When B is invertible, the problem is reduced to $B^{-1}A\mathbf{v} = \lambda\mathbf{v}$. However, explicitly forming $B^{-1}A$ is numerically unstable if B is ill-conditioned. Since B is symmetric and positive definite, one can compute a Cholesky factorization $B = LL^T$ which allows us to

reduce the equation to a standard eigenvalue problem $L^{-1}AL^{-T}\mathbf{y} = \lambda\mathbf{y}$, where $\mathbf{y} = L^T\mathbf{v}$, which can then be solved with the symmetric QR algorithm to compute a diagonal eigenvalue decomposition. A detailed treatment for this case is presented in [3]. In practice, this approach usually results in small relative residuals for eigenvalues that are large in magnitude and larger relative residuals for eigenvalues that are smaller in magnitude.

The QZ algorithm [5] for the non-symmetric GEP, is an iterative method that generalizes the QR algorithm to handle directly the generalized eigenvalue problem instead of the standard eigenvalue problem, thereby avoiding potential problems with inverting ill-conditioned or singular B . It applies orthogonal transformations to simultaneously reduce A and B to upper triangular forms from which the eigenvalues are extracted. Although this method is robust and backward stable, it is computationally expensive, thereby limiting its use to small or medium sized matrices.

A more stable approach for a dense symmetric semidefinite problem where B is symmetric positive definite and possibly ill-conditioned is described in [6]. This approach uses a spectral transformation method that leverages a shifted and inverted formulation of the problem, which it then solves using symmetric factorizations. Using a rook pivoted LDL^T factorization for $A - \sigma B$ and a Cholesky factorization for B , such that $A - \sigma B = C_a D_a C_a^T$ and $B = C_b C_b^T$, the generalized problem is transformed to a standard one given by

$$C_b^T C_a^{-T} D_a C_a^{-1} C_b \mathbf{u} = \theta \mathbf{u}. \quad (1.1)$$

Under some technical assumptions that typically hold in practice, it was proved that if

the scaled shift

$$\sigma_0 = \sigma \frac{\|B\|}{\|A\|} \quad (1.2)$$

is not too large and the shift σ is not chosen to be too close to a generalized eigenvalue, then this approach is stable in the sense that it gives generalized eigenvalues of a pair of matrices close to A and B . This is true regardless of any ill conditioning in $A - \sigma B$. This approach was validated by numerical experiments, contrasting it with Cholesky-based methods that are unstable for small eigenvalues.

1.3 Mathematical Preliminaries

In this section, we shall introduce some notations and the key mathematical concepts underlying the eigenvalue problems that will be used throughout this study.

1.3.1 Notation

Throughout this study, we make use of the following notations:

$A \in \mathbb{R}^{m \times n}$: denotes a matrix

$[A]_{ij}$: denotes element (i, j) of A

$\mathbf{x} \in \mathbb{R}^m$: denotes a column vector

A^T : denotes the transpose of matrix A

$\|\cdot\|$: denotes a vector or matrix norm

$A_{i:i',j:j'}$: denotes the $(i' - i + 1) \times (j' - j + 1)$ submatrix of A

1.3.2 Floating Point Arithmetic

We define a *floating point* number system, $\mathbf{F}$ as a bounded subset of the real numbers $\mathbb{R}$, such that the elements of $\mathbf{F}$ are the number 0 together with all numbers of the form

$$x = \pm(m/\beta^t)\beta^e,$$

where m is an integer in the range $1 \leq m \leq \beta^t$ known as the significand, $\beta \geq 2$ is known as the *base* or *radix* (typically 2), e is an arbitrary integer known as the exponent and $t \geq 1$ is known as the precision.

To ensure that a nonzero element $x \in \mathbf{F}$ is unique, we can restrict the range of $\mathbf{F}$ to $\beta^{t-1} \leq m \leq \beta^t - 1$. The quantity $\pm(m/\beta^t)$ is then known as the *fraction* or *mantissa* of x . We define the number $u := \frac{1}{2}\beta^{1-t}$ as the *unit roundoff* or *machine epsilon*. In a relative sense, the *unit roundoff* is as large as the gaps between floating point numbers get.

Let $fl : \mathbb{R} \rightarrow \mathbf{F}$ be a function that gives the closest floating point approximation to a real number, then the following theorem gives a property of the unit roundoff.

Theorem 1.3.1. *If $x \in \mathbb{R}$ is in the range of $\mathbf{F}$, then $\exists \epsilon$ with $|\epsilon| \leq u$ such that $fl(x) = x(1 + \epsilon)$.*

One way we could think of this is that, the difference between a real number and its closest floating point approximation is always smaller than u in relative terms.

1.3.3 Conditioning and Stability

Given a mathematical problem $f : X \rightarrow Y$, the conditioning of the problem refers to its inherent sensitivity to perturbations in the input, while stability of an algorithm pertains to the propagation of errors (such as rounding or approximation errors) by the algorithm when solving the problem computationally. A *well-conditioned* problem is one with the property that small perturbations of the input lead to only small changes in the output. An *ill-conditioned* problem is one with the property that small perturbations in the input leads to a large change in the output.

For any mathematical problem, we can associate a number called the *condition number* to that problem that tells us how well-conditioned or ill-conditioned the problem is. For the purpose of this thesis, we shall only be considering the condition number of matrices. Since matrices can be viewed as linear transformations from one vector space to another, it makes sense to define a condition number for matrices.

For a matrix $A \in \mathbb{R}^{m \times n}$, the condition number with respect to a given norm is defined as

$$\kappa(A) = \|A\| \cdot \|A^{-1}\|. \quad (1.3)$$

In simpler terms, the condition number quantifies how the relative error in the solution of a linear system $A\mathbf{x} = \mathbf{b}$ can be amplified when there is a small perturbation in the input vector $\mathbf{x}$. If $\kappa(A)$ is small, A is said to be *well-conditioned*; if $\kappa(A)$ is large, then A is said to be *ill-conditioned*. It should be noted that the notion of being “small” or “large” depends on the application or problem we are solving. If $\|\cdot\| = \|\cdot\|_2$ (spectral norm or 2-norm), then

$\|A\| = \sigma_1$ and $\|A^{-1}\| = 1/\sigma_m$, so that

$$\kappa(A) = \frac{\sigma_1}{\sigma_m}, \quad (1.4)$$

where σ_1 and σ_m are the largest and smallest singular values of A respectively. Throughout the remainder of this thesis, unless stated otherwise, $\|\cdot\|$ will refer to the spectral norm, or 2-norm.

1.3.4 The Generalized Eigenvalue Problem

Let $A, B \in \mathbb{R}^{m \times m}$ be arbitrary square matrices. A *matrix pencil* is an expression of the form $A - \lambda B$, where $\lambda \in \mathbb{C}$. The *eigenvalues* of the pencil $A - \lambda B$ are the elements of the set $\Lambda(A, B)$ defined by

$$\Lambda(A, B) = \{z \in \mathbb{C} : \det(A - zB) = 0\}. \quad (1.5)$$

That is, the generalized eigenvalues of A and B are precisely the roots of the characteristic polynomial of the matrix pencil, given by

$$p_{A,B}(\lambda) = \det(A - \lambda B) = 0. \quad (1.6)$$

The pencil $A - \lambda B$ is said to be *regular* if there exists at least one value of $\lambda \in \mathbb{C}$ such that $\det(A - \lambda B) \neq 0$, otherwise it is called *singular*. In other words, a pencil is singular if $\det(A - \lambda B) = 0$ for all $\lambda \in \mathbb{C}$. In this thesis, we shall assume that $A - \lambda B$ is regular.

If $\lambda \in \Lambda(A, B)$ and $\mathbf{0} \neq \mathbf{v} \in \mathbb{C}^m$ satisfies

$$A\mathbf{v} = \lambda B\mathbf{v}, \quad (1.7)$$

then $\mathbf{v}$ is a generalized eigenvector of A and B corresponding to λ . The problem of finding non-trivial solutions to (1.7) is known as the *generalized eigenvalue problem*.

If B is non-singular, then the problem reduces to a standard eigenvalue problem

$$B^{-1}A\mathbf{v} = \lambda\mathbf{v}. \quad (1.8)$$

In this case, the generalized eigenvalue problem has m eigenvalues if and only if $\text{rank}(B) = m$. This implies that the generalized eigenvalues of A and B are equal to the eigenvalues of $B^{-1}A$. If B is singular or rank deficient, this reduction is not possible, and the set of generalized eigenvalues $\Lambda(A, B)$ may be finite, empty or infinite. If $\Lambda(A, B)$ is finite, the number of eigenvalues will be less than m . This is because the characteristic polynomial $\det(A - \lambda B)$ is of degree less than m , so that there is not a complete set of eigenvalues for the problem.

If A and B have a nontrivial common null space, then every choice of λ will be a solution to (1.7) for some choice of $\mathbf{v}$. If $\mathbf{v}$ is in the null space of B , then we identify it as a generalized eigenvector corresponding to the generalized eigenvalue $\lambda = \infty$. For the purpose of this study, we shall assume that A and B do not have a nontrivial common null space, that is

$$\mathcal{N}(A) \cap \mathcal{N}(B) = \{\mathbf{0}\}. \quad (1.9)$$

When A and B are symmetric and B is positive definite, we shall call the problem symmetric-definite generalized eigenvalue problem, which will be the focus of this thesis. The symmetric-definite generalized eigenvalue problem is fully analogous to the symmetric standard eigenvalue problem. In this case we will see that the pencil is regular, the generalized eigenvalues are all real, and the generalized eigenvectors are orthogonal with respect to the inner product

$$\langle \mathbf{x}, \mathbf{y} \rangle_B = \mathbf{y}^T B \mathbf{x}.$$

1.3.5 Generalized Eigenvalue Problem with Symmetry

In this section, we shall consider the symmetric version of the generalized eigenvalue problem that was described in Section 1.3.4, which will be the main focus of this thesis. We will also consider the methodological approach used in solving the problem, and discuss the challenges involved in solving these kind of problems.

The symmetric-definite generalized eigenvalue problem is formally given by:

$$A\mathbf{v} = \lambda B\mathbf{v}, \quad \mathbf{v} \neq 0, \quad (1.10)$$

where $A \in \mathbb{R}^{m \times m}$ is symmetric and $B \in \mathbb{R}^{m \times m}$ is symmetric positive definite.

Problem (1.10) can be reformulated as

$$\beta A\mathbf{v} = \alpha B\mathbf{v}, \quad \mathbf{v} \neq 0. \quad (1.11)$$

We have replaced λ with α/β for convenience so that the generalized eigenvalues will be of the form (α, β) . If $\beta = 0$, then the infinite generalized eigenvalues will have the form $(\alpha, 0)$. The formulation using (1.11) is useful when describing the error bounds, as we shall later see. We shall alternate between (1.10) and (1.11) when convenient.

It is known that the eigenvalues of a symmetric definite generalized eigenvalue problem are real. An interesting property of the symmetric-definite problem is that it can be transformed to an equivalent symmetric definite problem with a congruence transformation. Let P be a

non-singular matrix, then (1.10), is equivalent to the transformed problem

$$(P^T AP)\mathbf{v} = \lambda(P^T BP)\mathbf{v}, \quad (1.12)$$

so that

$$\Lambda(A, B) = \Lambda(P^T AP, P^T BP). \quad (1.13)$$

Furthermore, any symmetric-definite pair (A, B) , A and B can be simultaneously diagonalized by a non-singular matrix P such that

$$P^T AP = D_a \quad \text{and} \quad P^T BP = D_b, \quad (1.14)$$

where $D_a = \text{diag}(a_1, \dots, a_n)$, and $D_b = \text{diag}(b_1, \dots, b_n)$. The generalized eigenvalues λ of A and B will be the diagonal elements of $D_b^{-1}D_a$, or put simply $\lambda_i = a_i/b_i$, and the eigenvectors will be the columns of P . The existence of P when B is semidefinite is given in [3, p. 498].

To compute the set of generalized eigenvalues $\Lambda(A, B)$ that satisfies (1.10), our approach in this thesis, similar to what was used in [6] and [1], is to transform the problem into a standard eigenvalue problem using a spectral transformation, after which we apply an iterative algorithm, like the Lanczos algorithm to compute the eigenvalues. In practice, we often compute a subset of these generalized eigenvalues corresponding to those in the vicinity of a given shift σ . To have a deep understanding of this approach, the next section discusses this approach in detail, focusing on the relationship between the eigenvalues of the original problem and the eigenvalues of the transformed problem.

1.3.6 Spectral Transformation

Spectral transformation in numerical linear algebra is a technique that is used to modify the spectrum of matrix in a controlled way. This is usually done to improve the convergence properties of an algorithm or to make certain matrix properties more accessible. In the context of eigenvalue problems, spectral transformation is often used in direct and iterative methods, where manipulating the matrix can help focus on certain eigenvalues or improve numerical stability.

The central idea behind spectral transformation is that by applying a rational or polynomial transformation to the matrix A , we can manipulate its eigenvalues to increase the magnitude of the eigenvalues we are interested in without changing their eigenvectors. There are various types of spectral transformation, but the one of particular interest in this thesis is the *shift-invert* transformation. The shift-invert transformation involves transforming the original problem into a shifted and inverted one which can then be solved using a direct or iterative solver. This method focuses on finding the eigenvalues near a specified shift σ . It is useful when one is interested in a few eigenvalues near a given point in the spectrum.

Consider the problem of computing the eigenvalues of a matrix $A \in \mathbb{R}^{m \times m}$. Assume that m is so large that computing all the eigenvalues of A is not computationally feasible but rather, we are interested in computing the eigenvalues in a certain region of the spectrum of A . We can pick a shift $\sigma \in \mathbb{R}$ that is not an eigenvalue of A to construct a new matrix by using a shift-invert transformation. The shifted and inverted matrix is then given by $(A - \sigma I)^{-1}$. The eigenvectors of $(A - \sigma I)^{-1}$ are the same as the eigenvectors of A , and

the corresponding eigenvalues are $(\lambda_j - \sigma)^{-1}$, for each eigenvalue λ_j of A . This shifts the spectrum of A , making the eigenvalues near σ much more prominent in the transformed matrix. This shifting strategy can be used in other iterative algorithms like the inverse iteration [7].

For the generalized eigenvalue problem given in (1.10), if we introduce a shift $\sigma \in \mathbb{R}$ so that $A - \sigma B$ is non singular, a simple shifted and inverted formulation of the problem is given by

$$(A - \sigma B)^{-1} B \mathbf{v} = \theta \mathbf{v}, \quad (1.15)$$

where $\theta = 1/(\lambda - \sigma)$. Note, however, that this formulation does not result in a symmetric standard eigenvalue problem.

Suppose σ is close enough to a generalized eigenvalue $\lambda_J \in \Lambda(A, B)$ much more than the other generalized eigenvalues, then $(\lambda_J - \sigma)^{-1}$ may be much larger than $(\lambda_j - \sigma)^{-1}$ for all $j \neq J$. This transformation will map the eigenvalues in the neighborhood of σ to the extreme part of the new spectrum, which can be favorable for the convergence of Krylov methods. However there is a problem. The formulation (1.15) is not symmetric, so we cannot use the Lanczos algorithm. We now consider a shift-and-invert formulation that preserves symmetry. Since B is positive definite, we can compute a Cholesky factorization $B = C_b C_b^T$. Given this decomposition, we can have a formulation that preserves symmetry, guaranteed by the following lemma

Lemma 1.3.2. *Let $A - \sigma B$ be nonsingular and $B = C_b C_b^T$, where C_b is a lower triangular matrix. Assume that $\lambda \neq \infty$ and $\mathbf{v} \neq \mathbf{0}$ satisfies (1.10). Then $\theta = 1/(\lambda - \sigma)$ is an eigenvalue*

of the problem

$$C_b^T(A - \sigma B)^{-1}C_b \mathbf{u} = \theta \mathbf{u}, \quad \mathbf{u} \neq \mathbf{0}, \quad (1.16)$$

with eigenvector $\mathbf{u} = C_b^T \mathbf{v} \neq \mathbf{0}$.

Conversely, assume that $\mathbf{u} \neq \mathbf{0}$ is an eigenvector for (1.16), with eigenvalue θ . If $C_b \mathbf{u} \neq \mathbf{0}$, then the eigenvector $\mathbf{v} = (A - \sigma B)^{-1}C_b \mathbf{u} \neq \mathbf{0}$ is an eigenvector for (1.11) with eigenvalue $(1 + \sigma\theta, \theta)$. In this case, with $\mathbf{v}$ defined in this way, we have $C_b^T \mathbf{v} = \theta \mathbf{u}$. If instead we have $C_b \mathbf{u} = \mathbf{0}$, then $\theta = 0$ and $(1, 0)$ is an eigenvalue for (1.11) with eigenvector given by $\mathbf{v} = \mathbf{u}$. If C_b is $m \times m$ and invertible, then we have $C_b \mathbf{u} \neq \mathbf{0}$ and can use the alternative formula $\mathbf{v} = C_b^{-T} \mathbf{u}$ to obtain an eigenvector of (1.11).

The proof of this lemma can be found in [6], although most claims of the lemma were used in [1]. In essence, Lemma 1.3.2 describes the relationship between the eigenvalues (*resp.* eigenvectors) of the original problem (1.10) and the eigenvalues (*resp.* eigenvectors) of the spectral transformed problem (1.16). Equation (1.16) gives us the spectral transformed version of the original generalized problem that preserves symmetry. Since the problem is now in a standard form, we can then apply the Lanczos algorithm to compute the desired eigenvalues within the neighborhood of σ , together with their corresponding eigenvectors. It should be noted that forming the spectral matrix in (1.16) is not desirable in a realistic problem since it does not preserve sparsity and will be very inefficient on most realistic problems. In practice, we employ stable factorizations of $(A - \sigma B)^{-1}$ that preserves symmetry in order to solve (1.16) effectively. Our choice of factorizations will be discussed in later chapters. For now, we turn our attention to the Krylov subspace method that will be

employed in solving (1.16).

1.3.7 Lanczos Algorithm

The Lanczos algorithm is an iterative method in numerical linear algebra used in finding the eigenvalues and eigenvectors of a *symmetric* matrix. It is particularly useful when dealing with large scale problems, where directly computing the eigenvalues and eigenvectors of the matrix would be computationally expensive and infeasible to compute in practice. It works by finding the “most useful” eigenvalues of the matrix — typically those at the extreme of the spectrum, and their eigenvectors. At its core, the main goal of the algorithm is to approximate the extreme eigenvalues and eigenvectors of a large, sparse, symmetric matrix by transforming the matrix into a smaller tridiagonal matrix that preserves the extremal spectral properties of the original matrix. This reduction is achieved by iteratively constructing an orthonormal basis of the Krylov subspace associated with the matrix.

Given a symmetric matrix $A \in \mathbb{R}^{m \times m}$, and an initial vector $\mathbf{v}_1$, the Lanczos algorithm produces a sequence of vectors $\mathbf{q}_1, \mathbf{q}_2, \dots, \mathbf{q}_n$ (where n is the number of iterations) that forms an orthonormal basis for the n -dimensional Krylov subspace

$$\mathcal{K}_n(A, \mathbf{v}_1) = \text{span}(\{\mathbf{v}_1, A\mathbf{v}_1, A^2\mathbf{v}_1, \dots, A^{n-1}\mathbf{v}_1\}). \quad (1.17)$$

This orthonormal basis is used to form a tridiagonal matrix T_n whose eigenvalues approximate the eigenvalues of A .

1.4 Motivation of Study

This study is motivated by several key factors that underscore the importance of advancing our understanding and capabilities in solving these type of problems. Originally, the motivation for this study arises from the need to compare the efficiency, accuracy and stability of iterative and direct methods for solving eigenvalue problems. In particular, the proven error bounds for the direct method in [6], shows that for a shift of moderate size that is not close to a generalized eigenvalue, the relative residuals are small for generalized eigenvalues that are not much larger than the shift. It is natural to ask if the same can be said for an iterative method like the Lanczos algorithm.

This study also aims to advance the field of numerical linear algebra. The insights gained from analyzing the spectral transformation Lanczos (ST-Lanczos) algorithm for generalized eigenvalue problems may inform the development of new algorithms or hybrid methods that combine the strengths of different methods. This could potentially lead to breakthroughs in the development of eigenvalue algorithms that are more reliable than the current ones we have today.

CHAPTER 2

METHODOLOGY AND ALGORITHM DESCRIPTION

In this chapter, we shall present a detailed description of the methodologies and implementation of algorithms used in this thesis to solve the generalized eigenvalue problem. We begin by describing the problem setup, followed by a discussion of the algorithms used, together with their implementation details. This chapter aims to provide a comprehensive understanding of how these algorithms are applied to derive the solutions to the problem at hand. We shall also give a description of the numerical experiments we setup to investigate the efficiency of these algorithms.

To compute the eigenvalues and eigenvectors that satisfy (1.10) with the spectral transformation Lanczos algorithm, our approach will be in two steps:

- Transform the generalized problem into a spectral transformed standard eigenvalue problem.
- Solve the spectral problem with the Lanczos algorithm.

As described in Section 1.3.6, Lemma 1.3.2 gives us the relationship between the eigenvalues (*resp.* eigenvectors) of the original problem and the eigenvalues (*resp.* eigenvectors) of the spectral transformed problem. Recall that solving (1.16) relies on symmetric factorization of the $A - \sigma B$. The following section describes the possible ways of doing this.

2.1 Stable Decompositions

We shall begin by computing decompositions for $A - \sigma B$ and B . If B is positive definite, we can compute a Cholesky decomposition $B = C_b C_b^T$ using the SciPy `cholesky` method which calls LAPACK `xPOTRF`. However, if B is positive semi-definite, this function call fails. One way to get around this is to use the pivoted Cholesky factorization provided by the `xPSTRF` routine. Although, SciPy does not expose `xPSTRF` directly through its `scipy.linalg` module, it can still be accessed via the lower-level `scipy.linalg.lapack` interface.

There are various possible factorization options for $A - \sigma B$. One option is to use the pivoted LDL^T factorization used by [6] and [1], where D is a block diagonal matrix with 1×1 and 2×2 blocks on the diagonal, and L is a lower triangular matrix. This factorization ideally uses the “rook pivoting” scheme, which is stable. Although the standard LDL^T factorization (without “rook pivoting”) is available in the SciPy Linear Algebra module, there is no option to use the rook pivoting scheme except if one chooses to write a custom LAPACK binding that makes use of `DSYTRF_ROOK`. While this can guarantee some stability for the problem we are trying to solve, it usually involves extra work in processing the 2×2 blocks to make D diagonal.

Another possible choice of factorization is an eigenvalue decomposition of $A - \sigma B$. If we use a symmetric eigenvalue decomposition $A - \sigma B = W D W^T$, our numerical experiments reveals that this stabilizes the Ritz residuals and the generalized residuals together with the advantage that these residuals are insensitive to the ill-conditioning in $A - \sigma B$. This can be done using built-in eigenvalue solvers in SciPy or any linear algebra library. This is the

most promising factorization, however computing eigenvalue decompositions for large sparse problems become computationally expensive and not feasible in practice. So, although the eigenvalue decomposition allows tests for stability, in a real implementation it would be replaced by an LDL^T factorization.

Lastly, we consider using an LU factorization for $A - \sigma B$. This approach is computationally less expensive and easier to implement. However, unlike the symmetric factorizations discussed earlier, the stability of the Ritz residuals is generally weaker, as it depends on the conditioning of $A - \sigma B$. Nevertheless, for the purpose of this thesis, we shall be comparing the LU factorization with the eigenvalue decomposition of $A - \sigma B$. This comparison helps to highlight a key source of instability that arises when symmetry is not preserved in the spectral shift.

One major takeaway from our experiments with the various options of factorizing $A - \sigma B$ is that symmetry is clearly important for stability. We plan to give a mathematical justification for this in future work.

Given our recent results, one might suggest using a stable decomposition such as the LU factorization. However, as we will demonstrate, decompositions that preserve symmetry exhibit certain stability advantages in practice.

2.2 The Lanczos decomposition

In this section, we revisit the Lanczos algorithm, and discuss how we apply it to the spectral transformed problem. As discussed in Section 1.3.7, the Lanczos algorithm approximates

the eigenvalues of the original problem by projecting it onto a Krylov subspace spanned by successive powers of the system matrix applied to an initial vector. The eigenvalue approximations arises from the tridiagonal matrix obtained through the Lanczos process, which captures the essential spectral characteristics of the original matrix.

Given $A \in \mathbb{R}^{m \times m}$, with $A = A^T$, the pseudocode for the Lanczos algorithm is described by Algorithm 1. After the completion of Algorithm 1, the γ_j and δ_j are used to construct

Algorithm 1 Lanczos Algorithm for a Symmetric Matrix

Require: $A = A^T$, number of iterations: n , tolerance: tol

```

1: function LANCZOS( $A, n, tol$ )
2:   Choose arbitrary  $\mathbf{b}$ ; set  $\mathbf{q}_1 = \mathbf{b}/\|\mathbf{b}\|_2$ 
3:   Initialize  $Q_n = [\mathbf{q}_1]$ ,  $\mathbf{q}_0 = \mathbf{0}$ ,  $\delta_0 = 0$ 
4:   Initialize  $\gamma = []$ ,  $\delta = []$  ▷ Arrays for  $T_n$  entries
5:   for  $j = 1, 2, \dots, n$  do
6:      $\mathbf{v} = A\mathbf{q}_j$  ▷ Matrix-vector product
7:      $\gamma_j = \mathbf{q}_j^T \mathbf{v}$  ▷ Diagonal entry  $\gamma_j$ 
8:      $\mathbf{v} = \mathbf{v} - \delta_{j-1}\mathbf{q}_{j-1} - \gamma_j\mathbf{q}_j$  ▷ 3-term recurrence
9:      $\mathbf{v} = \mathbf{v} - \sum_{i=1}^j (\mathbf{q}_i^T \mathbf{v})\mathbf{q}_i$  ▷ Full reorthogonalization
10:     $\delta_j = \|\mathbf{v}\|_2$  ▷ Off-diagonal entry  $\delta_j$ 
11:    if  $\delta_j < tol$  then
12:      break ▷ Handle breakdown
13:    end if
14:     $\mathbf{q}_{j+1} = \mathbf{v}/\delta_j$ 
15:    Append  $\mathbf{q}_{j+1}$  to  $Q_n$  and  $\gamma_j, \delta_j$  to  $\gamma, \delta$ 
16:  end for
17:  Truncate  $Q_n$  to first  $n$  vectors:  $Q_n = [\mathbf{q}_1, \dots, \mathbf{q}_n]$ 
18:  return  $Q_n, \gamma, \delta, \mathbf{q}_{n+1}$ 
19: end function

```

the tridiagonal matrix $T_n \in \mathbb{R}^{n \times n}$ and the vectors $\mathbf{q}_j$ are stacked together to form a matrix

$Q_n \in \mathbb{R}^{m \times n}$ given by:

$$T_n = \begin{pmatrix} \gamma_1 & \delta_1 & & & \\ \delta_1 & \gamma_2 & \delta_2 & & \\ & \delta_2 & \gamma_3 & \delta_3 & \\ & & \ddots & \ddots & \vdots \\ & & & \delta_{n-1} & \gamma_n \end{pmatrix}$$

$$Q_n = \left[\begin{array}{c|c|c|c} \mathbf{q}_1 & & & \\ \hline & \mathbf{q}_2 & & \\ \hline & & \cdots & \\ \hline & & & \mathbf{q}_n \end{array} \right].$$

The decomposition is given by

$$AQ_n = Q_n T_n + \delta_n \mathbf{q}_{n+1} \mathbf{e}_n^T. \quad (2.1)$$

In theory, the vectors $\mathbf{q}_j$ should be orthonormal, but due to floating-point errors, there will be loss of orthogonality, hence the need for line 9 in Algorithm 1.

Let $\theta_i, i = 1, 2, \dots, n$ (which can be computed by standard eigenvalue solvers) be the eigenvalues of T_n , and $\{\mathbf{y}_i\}_{i=1:n}$ be the associated eigenvectors. The $\{\theta_i\}$ are called the *Ritz values* and the vectors $\{Q_n \mathbf{y}_i\}_{i=1:n}$ are called the *Ritz vectors*. The extreme eigenvalues of A are typically well approximated by the extreme Ritz values derived from the Lanczos process, with the corresponding Ritz vectors serving as approximations to A 's associated eigenvectors.

Since the generalized eigenvalue problem we started with has been reduced to a standard one as shown in equation (1.16), Algorithm 1 can be applied to equation (1.16) with some slight modifications. The spectral form of Algorithm 1 is given by Algorithm 2. After applying the Lanczos procedure to the spectral transformed problem (1.16), we compute the

Algorithm 2 Spectral Lanczos Algorithm for (1.16)

Require: $A = A^T$, $B = B^T$, with B being positive definite or semidefinite

Require: number of iterations: n , size of matrix A or B : m , tolerance: tol

Require: $\sigma \in \mathbb{R}$: shift not close to a generalized eigenvalue

```
1: function SPECTRAL_LANCZOS( $A, B, n, \sigma, tol$ )
2:   Choose an arbitrary vector  $\mathbf{b}$ ; set  $\mathbf{q}_1 = \mathbf{b}/\|\mathbf{b}\|_2$ 
3:   Initialize  $Q_n = [\mathbf{q}_1]$ ,  $\mathbf{q}_0 = \mathbf{0}$ ,  $\delta_0 = 0$ 
4:   Initialize  $\gamma = [], \delta = []$  ▷ Arrays for  $T_n$  entries
5:    $LU = (A - \sigma B)$  ▷ Precompute the  $LU$  factorization of shifted system
6:   Factor:  $B = CC^T$  ▷ Cholesky factorization
7:   for  $j = 1, 2, \dots, n$  do
8:      $\mathbf{u} = C\mathbf{q}_j$  ▷ Matrix vector product
9:      $\mathbf{v} = \text{LINEARSOLVE}(LU, \mathbf{u})$  ▷ Forward/Backward substitution
10:     $\mathbf{v} = C^T \mathbf{v}$  ▷ Matrix-vector product
11:     $\gamma_j = \mathbf{q}_j^T \mathbf{v}$  ▷ Diagonal entry  $\gamma_j$ 
12:     $\mathbf{v} = \mathbf{v} - \delta_{j-1} \mathbf{q}_{j-1} - \gamma_j \mathbf{q}_j$  ▷ 3-term recurrence
13:     $\mathbf{v} = \mathbf{v} - \sum_{i=1}^j (\mathbf{q}_i^T \mathbf{v}) \mathbf{q}_i$  ▷ Full reorthogonalization
14:     $\delta_j = \|\mathbf{v}\|_2$  ▷ Off-diagonal entry  $\delta_j$ 
15:    if  $\delta_j < tol$  then
16:      break
17:    end if
18:     $\mathbf{q}_{j+1} := \mathbf{v}/\delta_j$ 
19:    Append  $\mathbf{q}_{j+1}$  to  $Q_n$  and  $\gamma_j, \delta_j$  to  $\gamma, \delta$ 
20:  end for
21:  Truncate  $Q_n$  to first  $n$  vectors:  $Q_n = [\mathbf{q}_1, \dots, \mathbf{q}_n]$ 
22:  return  $Q_n, \gamma, \delta, \mathbf{q}_{n+1}$ 
23: end function
```

converged Ritz pairs using a certain tolerance. The converged Ritz pairs are then mapped to the generalized eigenvalues and eigenvectors where we can observe the behaviour of their residuals with respect to conditioning.

2.3 Problem Setup

To evaluate the performance and robustness of the spectral transformation Lanczos algorithm, we set up a problem with predetermined eigenvalues, use the algorithm to compute

the eigenvalues, and show that the residuals follow closely with the bounds predicted for direct methods in [6]. While there are other options of using matrices from open source repositories like Matrix Market¹, we choose to use this approach so that we can control the size, condition number and other properties of the matrix so as to observe the effect of this properties on the algorithm.

Starting with a diagonal matrix $D \in \mathbb{R}^{m \times m}$ with known eigenvalues, we generate a random matrix P of size $m \times m$ with standard normal distribution. Since the QR factorization is guaranteed to exist for any matrix, we take the QR factorization of P to obtain an orthogonal matrix Q , which is used to create a matrix C using orthogonal transformation. Hence $C = QDQ^T$ is unitarily similar to D .

Next, we initialize a random lower triangular matrix $L_0 \in \mathbb{R}^{m \times m}$ with a normal distribution. A symmetric positive definite $B \in \mathbb{R}^{m \times m}$ is formed by

$$B = L_0 L_0^T + \delta I_m, \quad \delta > 0, \quad (2.2)$$

where I_m is an identity matrix of order m . Clearly, B is symmetric. The matrix $L_0 L_0^T$ is positive semi-definite since for any non-zero vector $\mathbf{x}$

$$\mathbf{x}^T (L_0 L_0^T) \mathbf{x} = (L_0^T \mathbf{x})^T (L_0^T \mathbf{x}) = \|L_0^T \mathbf{x}\|^2 \geq 0. \quad (2.3)$$

However, $L_0 L_0^T$ may not be strictly positive definite if L_0 is singular. The term δI_m ensures strict positive definiteness by adding δ to its diagonals, thereby shifting all eigenvalues by δ .

¹<https://math.nist.gov/MatrixMarket/index.html>

If $\delta > 0$, then all eigenvalues of B will be strictly positive, ensuring B is positive definite. This guarantees that we can compute the Cholesky factorization of B without any numerical issues.

Another important thing to note is that δ can be used to control the conditioning of B . We recall from Section 1.3.3, that the condition number of B if $B = B^T$, is defined as:

$$\kappa(B) = \frac{\lambda_{\max}(B)}{\lambda_{\min}(B)}, \quad (2.4)$$

where $\lambda_{\max}(B)$ and $\lambda_{\min}(B)$ are the largest and smallest eigenvalues of B , respectively. In general, B is usually ill-conditioned with a very large condition number so that if δ is large, the process of adding δI_m can regularize the condition number of B , making B well-conditioned, since that will equate to increasing $\lambda_{\min}(B)$. Even for small δ , B may remain numerically ill-conditioned, though not to pathological extremes. Hence, δ is a hyperparameter we can use to control the condition of B .

Since B is symmetric and positive definite, we can compute its Cholesky factorization $B = LL^T$ and construct A using a congruence transformation

$$A = LCL^T. \quad (2.5)$$

So that the generalized eigenvalues $\Lambda(A, B)$ are equal to the eigenvalues of the diagonal matrix D . This can be summarized by the following lemma:

Lemma 2.3.1. *Let $A - \lambda B$ be a pencil, where A and B are symmetric, and B is strictly positive definite. Let D be a diagonal matrix and C be unitarily similar to D . Assuming (2.5) holds, then the generalized eigenvalues in $\Lambda(A, B)$ are the diagonal elements of D .*

Proof. Given the generalized problem

$$A\mathbf{v} = \lambda B\mathbf{v}, \quad \mathbf{v} \neq \mathbf{0} \quad (2.6)$$

Since B is positive definite, then clearly, it is invertible and the generalized eigenvalues $\Lambda(A, B)$ will be the eigenvalues of $B^{-1}A$.

Now

$$\begin{aligned} B^{-1}A &= (LL^T)^{-1}(LCL^T) \\ &= L^{-T}L^{-1}LQDQ^TL^T \\ &= (L^{-T}Q)D(Q^{-1}L^T) \\ &= (L^{-T}Q)D(L^{-T}Q)^{-1} \end{aligned}$$

Therefore $B^{-1}A$ is similar to D . □

The pseudocode for generating A and B is described in Algorithm 3. With the problem

Algorithm 3 Setting up a GEP

Require: D : diagonal matrix with known eigenvalues, δ : regularization hyperparameter

```

1: function GENERATE_MATRIX( $D, \delta$ )
2:   Set  $m = \text{size}(D)$ 
3:    $Q, \_ = \text{qr}(\text{random.randn}(m, m))$ 
4:    $C = QDQ^T$ 
5:    $L_0 = \text{tril}(\text{random.randn}(m, m))$ 
6:    $B = (L_0L_0^T) + \delta I$ 
7:    $L = \text{cholesky}(B)$ 
8:    $A = LCL^T$ 
9:   return ( $A, B$ )
10: end function

```

setup completed, and the algorithm described, in the next chapter, we shall discuss the

results obtained in these experiments.

CHAPTER 3

EXPERIMENTAL RESULTS AND DISCUSSION

In this chapter, we will present a comprehensive analysis of the experimental results obtained from our implementation of the spectral transformation Lanczos algorithm for the symmetric definite generalized eigenvalue problem. We examine the algorithm performance on a test matrix, analyze the effects of ill-conditioning on convergence and accuracy, and compare the error bounds with what was predicted with direct methods.

3.1 Residual and Error Analysis

To evaluate the efficiency of the ST-Lanczos algorithm for the given problem, we define some metrics and also review some of the residual bounds theorems given in [6].

Definition 3.1.1 (Lanczos Decomposition Residual). *Let equation (2.1) be the decomposition obtained from the completion of Algorithm 2. Then we define the relative residual as:*

$$\text{Relative Decomposition Residual} = \frac{\|C_b^T(A - \sigma B)^{-1}C_b Q_n - Q_n T_n - \delta_n \mathbf{q}_{n+1} \mathbf{e}_n^T\|}{\|C_b^T(A - \sigma B)^{-1}C_b\|}. \quad (3.1)$$

This residual measures how well the Lanczos algorithm has approximated the eigenvalues of B through the tridiagonal matrix T_n .

Definition 3.1.2 (Generalized Relative Residual). *Let (α_i, β_i) be the generalized eigenvalues of A and B , and $\mathbf{v}_i \neq \mathbf{0}$ be the corresponding computed eigenvector obtained from computing the eigenvalues of T_n , such that $\mathbf{r}_i = (\beta_i A - \alpha_i B)\mathbf{v}_i$. Then we define the relative residual for*

each $i = 1, 2, \dots, n$ as:

$$\|\tilde{\mathbf{r}}_i\| = \frac{\|(\beta_i A - \alpha_i B)\mathbf{v}_i\|}{(|\beta_i|\|A\| + |\alpha_i|\|B\|)\|\mathbf{v}_i\|}. \quad (3.2)$$

Definition 3.1.3 (Spectral Transformed Residual). *Let $(\theta_i, \mathbf{u}_i)$ be the Ritz-pair obtained from computing the eigenvalues and eigenvectors of T_n . Then we define the relative residual for each $i = 1, 2, \dots, n$ as:*

$$ST \text{ Relative Residual} = \frac{\|C_b^T(A - \sigma B)^{-1}C_b\mathbf{u}_i - \theta_i\mathbf{u}_i\|}{(\|C_b^T(A - \sigma B)^{-1}C_b\| + |\theta_i|)\|\mathbf{u}_i\|}. \quad (3.3)$$

Definition 3.1.4 (Best Residuals). *Let (α_i, β_i) be the computed generalized eigenvalues of A and B . We define the “best relative residuals” for (α_i, β_i) as*

$$\|\tilde{\mathbf{r}}_i\| = \frac{\sigma_m(\beta_i A - \alpha_i B)}{(|\beta_i|\|A\| + |\alpha_i|\|B\|)}, \quad (3.4)$$

where σ_m is the smallest singular value of the matrix pencil $(\beta_i A - \alpha_i B)$.

This residual assess the quality of the computed eigenvalues independently of the eigenvectors. The smallest singular value is used to determine the “**best achievable residual**” for the eigenvalue pair (α_i, β_i) , even with an idealized eigenvector.

We reference the following theorem(s) and lemma(s) from [6], which gives the residual bounds for computed eigenvalues and eigenvectors using the direct method employed in the paper. We will be using these bounds to evaluate the efficiency and stability of the ST-Lanczos algorithm used in this thesis and validate our computed residuals with these bounds.

For context we define

$$\eta = \frac{\|A - \sigma B\|^{1/2}}{\|B\|^{1/2}}, \quad X = C_a^{-1}C_b, \quad \sigma_0 = \sigma \frac{\|B\|}{\|A\|}, \quad \text{and} \quad \mu = \frac{\|X\|}{\|X^T D_a X\|},$$

where σ_0 is known as the “*scaled-shift*”.

It is shown in [6] that

$$\eta^2 \|X\|_2^2 \leq \left(1 + \frac{1}{\sigma_0}\right) \frac{\mu}{\min_i \left|1 - \frac{\lambda_i}{\sigma}\right|}$$

where the minimum is taken over all generalized eigenvalues λ_i and σ_0 is as in (1.2). Hence, if μ is not large, if σ is not too close to a generalized eigenvalue, and if σ_0 is not too small, then $\eta^2 \|X\|_2^2$ is not large. These conditions are easily satisfied in practice, so that the presence of $\eta^2 \|X\|^2$ in error bounds poses no problem. It was also shown in [6] that if the spectral transformation matrix is computed explicitly and its eigenvalue decomposition is then computed, we obtain

$$(C_b + F_1)^T (A + E - \sigma(B + F))^{-1} (C_b + F_1) = \tilde{U}(\Theta + G)\tilde{U}^T, \quad (3.5)$$

$$B + F = (C_b + F_1)^T (C_b + F_1), \quad (3.6)$$

where Θ is the diagonal matrix of computed eigenvalues and $\tilde{U}$ is an exactly orthogonal matrix that is close to the matrix of computed eigenvectors for the spectral transformation matrix $C_b^T (A - \sigma B)^{-1} C_b$. The error matrices E , F , and G can be suitably bounded in terms of the unit roundoff. From these error relations it follows that the computed eigenvalues in Θ are close to the eigenvalues of the matrix obtained from a spectral transformation of matrices $A + E$ and $B + F$ that are close to A and B . These error bounds also have consequences for

the residuals associated with the computed solution to the generalized eigenvalue problem.

Theorem 3.1.1 (Residual Bounds for Eigenvalues). *Let C_a, C_b, X, η , and θ_i be computed quantities using the direct method algorithm described in [6]. Let E, F , and F_1 be as in (3.5) and (3.6). In what follows e_n and f_n are modestly growing functions of n more precisely described in [6]. Suppose that $(A + E) - \sigma(B + F)$ is invertible. Without loss of generality, assume that the computed eigenvalues θ_i are in nondecreasing order and let $\hat{\theta}_i$ and $\hat{\mathbf{u}}_i$ for $i = 1, 2, \dots, r$ be eigenvalues and eigenvectors of*

$$\hat{W} = (C_b + F_1)^T (A + E - \sigma(B + F))^{-1} (C_b + F_1)$$

with the $\hat{\theta}_i$ also in nondecreasing order. Assume that $(C_b + F_1)\hat{\mathbf{u}}_i \neq \mathbf{0}$ and define

$$\hat{\mathbf{v}}_i = (A + E - \sigma(B + F))^{-1} (C_b + F_1)\hat{\mathbf{u}}_i \neq \mathbf{0}.$$

Then

$$\|(\theta_i(A + E) - (1 + \sigma\theta_i)(B + F))\hat{\mathbf{v}}_i\|_2 \leq$$

$$u(e_n + f_n)(1 + |\sigma_0|)\eta^2 \|X\|_2^2 (|\theta_i| \|A + E\|_2 + |1 + \sigma\theta_i| \|B + F\|_2) \|\hat{\mathbf{v}}_i\|_2 + O(u^2).$$

where $\hat{\mathbf{v}}_i$ is the exact eigenvector of the perturbed problem, and not a computed eigenvector using the Lanczos procedure.

The implication of this theorem is if neither σ_0 nor $\eta\|X\|$ are large, then there exists an approximate eigenvector for which the computed eigenvalues achieve a small residual, although this eigenvector is not actually the computed eigenvector. Thus, the computation

of the eigenvalues on their own is stable. What is important for us to remember here is that choosing the shift σ to avoid extreme proximity to eigenvalues can be shown to give reasonable bounds on $\eta\|X\|_2$. That is, for eigenvalues $\lambda_i = (1 + \sigma\theta_i)/\theta_i$, the residuals will remain small if σ is not too close to λ_i .

The situation for the computed eigenvectors is not as satisfactory.

Theorem 3.1.2. *Assume that $\sigma \neq 0$ and $\lambda_i \neq 0$. Under the assumptions of Theorem 3.1.1, we have*

$$\begin{aligned} \|(\theta_i A - (1 + \sigma\theta_i)B)\mathbf{v}_i\|_2 \leq \\ |1 + \sigma\theta_i| \cdot |1 - \sigma/\lambda_i| \left[uc_n + O(u) \left(\eta\|X\|_2 \right. \right. \\ \left. \left. + \left(1 + \max(\gamma, 1) \left(1 + \left| 1 - \frac{\lambda_i}{\sigma} \right| \right) \right) \eta^2\|X\|_2^2 \right) \right] \|B\|_2 \|\mathbf{v}_i\|_2, \end{aligned}$$

and

$$\begin{aligned} \|(\theta_i A - (1 + \sigma\theta_i)B)\mathbf{v}_i\|_2 \leq \\ |\theta_i| \cdot |1 - \lambda_i/\sigma| \cdot |\sigma_0| \left[uc_n + O(u) \left(\eta\|X\|_2 \right. \right. \\ \left. \left. + \left(1 + \max(\gamma, 1) \left(1 + \left| 1 - \frac{\lambda_i}{\sigma} \right| \right) \right) \eta^2\|X\|_2^2 \right) \right] \|A\|_2 \|\mathbf{v}_i\|_2, \end{aligned}$$

where $\gamma = \|A\|_2/\|A - \sigma B\|_2$ and c_n does not grow quickly in n .

Theorem 3.1.2 covers two distinct cases. The first case is of primary interest when σ_0

is large. Again, assuming that σ is not so close to an eigenvalue that $\eta\|X\|$ becomes large, in this case the residuals scale with $|1 - \sigma/\lambda_i|$ and $|1 - \lambda_i/\sigma|$, ensuring accuracy when σ aligns somewhat closely with λ_i but suggesting a loss of accuracy if λ_i is either much larger or smaller than σ . The second case includes $|\sigma_0|$ as a factor in the bounds and is more useful when the scaled shift is not too large. In this case, the residuals scale with only with $|1 - \lambda_i/\sigma|$, so that we do not expect a loss of accuracy when λ_i is much smaller than σ_0 .

So the key takeaway useful for this thesis is, in order to ensure small residuals and stability, the shift σ should not be too close to any generalized eigenvalue and the scaled shift σ_0 should not be too large in magnitude. A moderate shift σ_0 will guarantee small residuals for eigenvalues not much larger than σ but a large shift will result in having small residuals only for eigenvalues near σ . The question that we will resolve in the experiments is whether this pattern holds for the Lanczos algorithm and not just for the direct method from [6].

3.2 Software and Computational Environment

The numerical experiments in this thesis are performed using the Python programming language together with the NumPy 2.0.2 and SciPy 1.13.1 libraries which makes function calls to optimized and efficient LAPACK and BLAS routines for linear algebra computations. These libraries ensure high-performance matrix operations and numerical stability. All computations are performed in **double precision** (64 bit floating point, `float64`) to maintain numerical accuracy and consistency.

For reproducibility, all code is written in Python 3.9.6 and executed within a controlled environment using `virtualenv`. All numerical results have been validated by comparing different levels of precision where applicable and verifying consistency with analytical results when available. Code for the experiments is managed using version control with Git to ensure reproducibility and can be found in this repository.¹

3.3 Experimental Setup

To evaluate the ST-Lanczos algorithm, we employ Algorithm 3 to generate test matrices A and B with controlled eigenvalue distribution. For the purpose of this thesis, we will be testing with dense matrices. In practice, the ST-Lanczos method is typically applied to sparse matrices. However, we will be experimenting with dense matrices, as they are easier to generate and more convenient for testing stability. The eigenvalues are divided into 3 groups, each with a specified range (spread). For each of the three groups, a random set of eigenvalues was generated with a uniform distribution, ensuring that the eigenvalues are distributed evenly within their respective ranges.

- Group 1 contains 1000 eigenvalues in the range $(10^{-3}, 10^1)$
- Group 2 contains 1000 eigenvalues in the range $(10^1, 10^3)$
- Group 3 contains 1000 eigenvalues in the range $(10^3, 10^7)$

The three sets of eigenvalues are then concatenated into a single array $D \in \mathbb{R}^{3000 \times 3000}$, which is then used with a regularization hyperparameter $\delta = 10^{-3}$, to generate A and B

¹<https://github.com/AyobamiAdebesin/ayobami.thesis>

of size 3000×3000 . We used a relatively moderate shift $\sigma_0 = 0.0017$ corresponding to $\sigma = 1.5 \times 10^3$. For our choice of δ , the system is relatively well-conditioned as shown below:

$$\kappa(A) = 4.25 \times 10^{15}, \quad \kappa(B) = 8.08 \times 10^6, \quad \kappa(A - \sigma B) = 5.5 \times 10^{12}.$$

As discussed in Section 2.3, A and B will be symmetric with B being positive definite. The matrix B is factored using the SciPy `cholesky` method which calls LAPACK `xPOTRF`. We chose to use this since B was designed to be strictly positive definite, which does not require us to use pivoted Cholesky factorization. If it were merely semidefinite, pivoting would be required. We run Algorithm 2 with $n = 2000$ iterations to solve the spectral problem

$$C_b^T (A - \sigma B)^{-1} C_b \mathbf{u} = \theta \mathbf{u}, \quad \mathbf{u} \neq \mathbf{0}, \quad (3.7)$$

and compute the Lanczos decomposition using the decompositions techniques for $A - \sigma B$ discussed in Section 2.1. We explore the results obtained from using these techniques and observe the behavior of the residual bounds in the following sections.

3.3.1 Eigenvalue Decomposition - Moderate Shift

We begin by considering the eigenvalue decomposition of $A - \sigma B$ given by

$$A - \sigma B = W D W^T, \quad (3.8)$$

so that the spectral transformed problem is given by

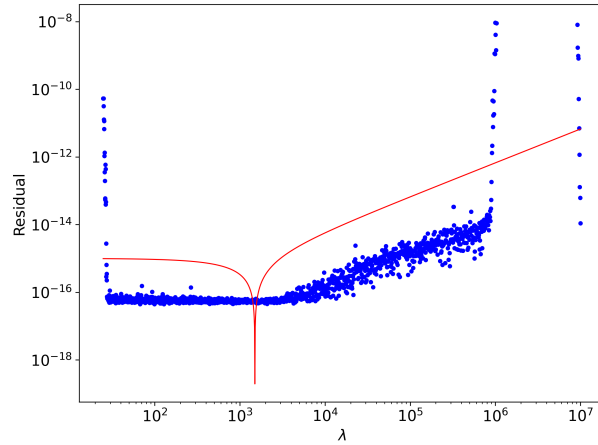
$$C_b^T W^{-T} D^{-1} W^{-1} C_b \mathbf{u} = \theta \mathbf{u}, \quad \mathbf{u} \neq \mathbf{0}. \quad (3.9)$$

This decomposition, done using the `linalg.eigh` function in SciPy, uses the LAPACK `dsyevd` routine for real symmetric matrices, which in turn uses divide-and-conquer algorithms for efficiency. For a moderate shift $\sigma_0 = 0.0017$ corresponding to $\sigma = 1.5 \times 10^3$, the Lanczos decomposition residual was observed to be of the order 10^{-27} , indicating a highly accurate Lanczos decomposition. As shown in Fig 3.1(a), using an eigenvalue decomposition achieves small residuals for eigenvalues close to or smaller in magnitude than the shift and gradually tends to increase for larger eigenvalues, consistent with the error bounds stated in Theorem 3.1.2. We observe that these residuals approximately scale with $10^{-15}|1 - \lambda_i/\sigma|$. This scale factor appears in one of the error bounds in Theorem 3.1.2.

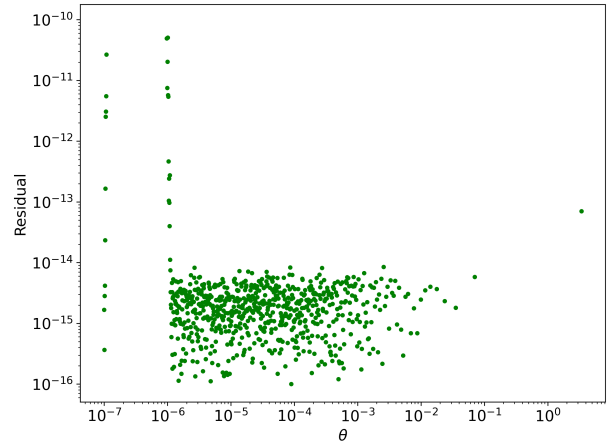
Fig 3.1(b) shows a plot of the Ritz residuals converging with an accuracy of machine precision. Fig 3.1(c) also shows the best possible relative residuals for some choice of $\mathbf{v}_i$. This shows that every computed eigenvalue can result in a small residual, validating the bound given in Theorem 3.1.1.

3.3.2 Eigenvalue Decomposition - Large Shift

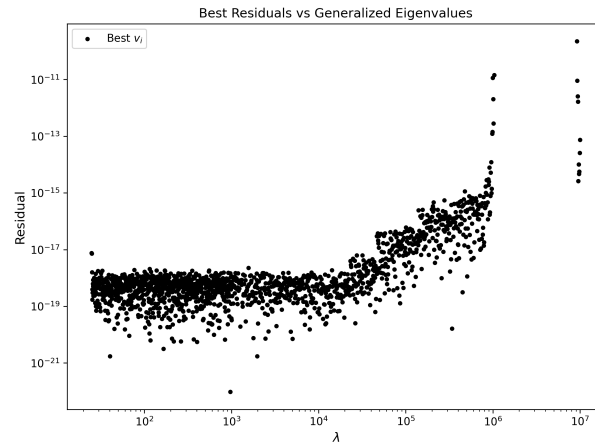
Next, we consider the effect of a large scaled shift on the eigenvalue decomposition. For a relatively large shift $\sigma_0 = 1.72$ corresponding to $\sigma = 1.5 \times 10^6$, we have that $\kappa(A - \sigma B) = 3.10 \times 10^8$. In this case, the Lanczos decomposition residual is 10^{-32} , also indicating a highly accurate Lanczos decomposition. We observe that eigenvalues with orders of magnitude smaller than the shift achieves large residuals, with those not too much smaller or larger in magnitude than the shift achieving small residuals as shown in Fig 3.2(a). The theoretical bound (red line) is $10^{-16}|(1 - \lambda_i/\sigma)(1 - \sigma/\lambda_i)|$, which appears as a factor in one of the error



(a) Residual vs Generalized Eigenvalues



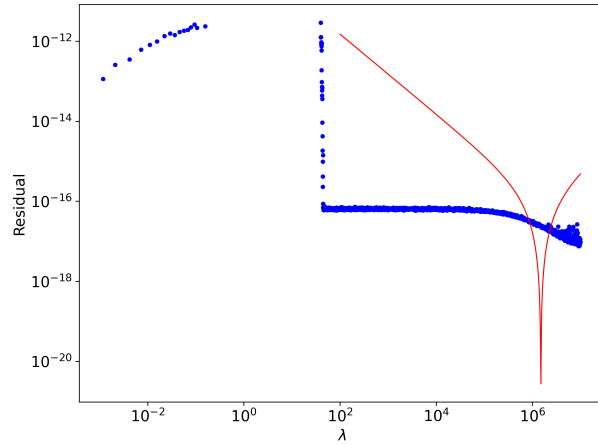
(b) Residuals vs Ritz values



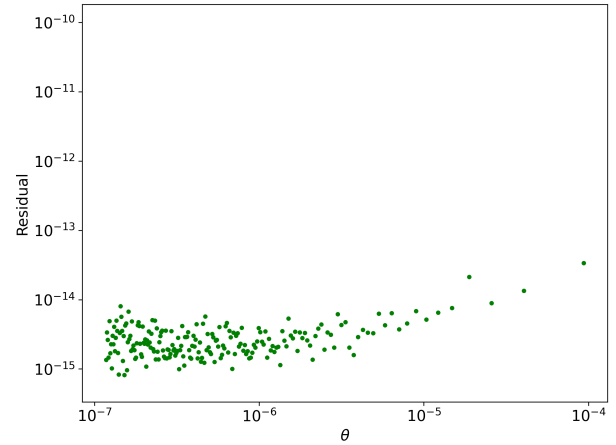
(c) Best Residuals

Figure 3.1: Residual plot with moderate shift

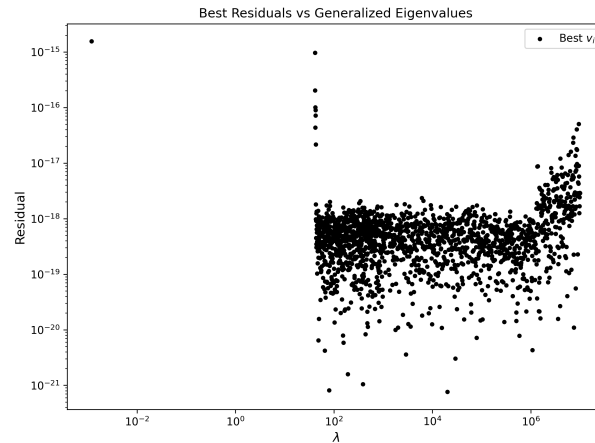
terms in the second bound in Theorem 3.1.2. This follows closely with what we expect from the error bounds. Infact, residuals for smaller eigenvalues do not appear to increase as much as the theoretical bound suggested. On the right, we have Fig 3.2(b) which shows a plot of the Ritz residuals converging with an accuracy of machine precision. Fig 3.1(c) shows that eigenvalues that are several orders of magnitude smaller than the shift cannot achieve a large residual except for those close to the shift.



(a) Residual vs Generalized Eigenvalues



(b) Residuals vs Ritz values



(c) Best Residuals

Figure 3.2: Residual plot with large shift

3.4 Effect of Ill-conditioning on Stability

As seen in the previous section, a symmetric decomposition for $A - \sigma B$ for the ST-Lanczos algorithm gave residuals that corresponds to the error bounds given in Theorem 3.1.2. However, the main advantage of the direct method given in [6] is that its stability is not affected by the ill-conditioning of $A - \sigma B$. In this section, we explore the effect of ill-conditioning for the symmetric and non-symmetric decomposition of $A - \sigma B$. To do this, we shall consider the

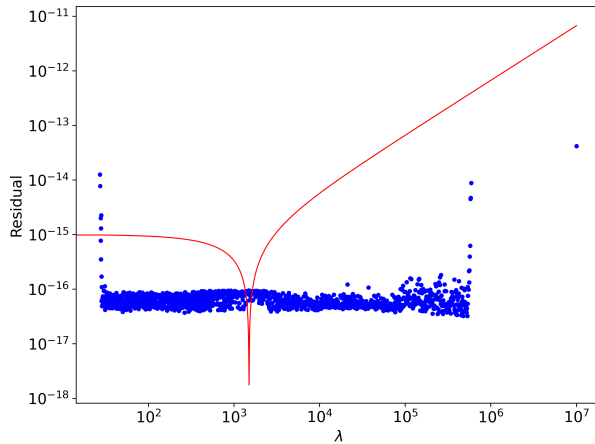
moderate scaled shift $\sigma_0 = 0.0017$ corresponding to $\sigma = 1.5 \times 10^3$, set $\delta = 10^{-12}$, and employ Algorithm 3 to generate new A and B , so that the system is even more ill-conditioned:

$$\kappa(A) = 4.17 \times 10^{18}, \quad \kappa(B) = 3.91 \times 10^{17}, \quad \kappa(A - \sigma B) = 3.75 \times 10^{18}.$$

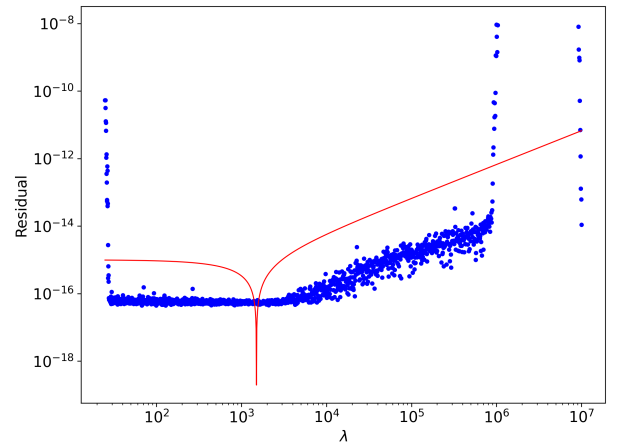
We will be comparing the residual bounds for this problem with the well-conditioned problem in Section 3.3.

3.4.1 Eigenvalue Decomposition

We ran Algorithm 2 with the eigenvalue decomposition of $A - \sigma B$ on the ill-conditioned problem. Again, the algorithm ran to completion with a Lanczos decomposition residual of 1.02×10^{-25} . As shown in Fig 3.3(a), the computed eigenvalues and eigenvectors still achieves a residual close to machine precision for the ill-conditioned problem, similar to what we have with the well-conditioned problem in Fig 3.3(b). This indicates that using a symmetric decomposition stabilizes the Lanczos process.



(a) Ill-Conditioned Problem



(b) Well-Conditioned Problem

Figure 3.3: Residual plot comparison with a moderate shift (Symmetric Decomposition)

3.4.2 *LU decomposition*

Next, we consider the effect of ill-conditioning on stability for a non-symmetric decomposition like the *LU* decomposition. The *LU* decomposition of $A - \sigma B$ is performed using the `linalg.lu_factor` function in SciPy, which employs partial pivoting. With a shift $\sigma = 1.5 \times 10^3$ not close to a generalized eigenvalue as before, we obtain small residuals for eigenvalues close to or smaller than the shift in magnitude in the case of the well-conditioned problem as shown in Fig 3.4(b). These small residuals are close to machine precision. The theoretical bound (red curve) is a plot of $10^{-12}|1 - \lambda_i/\sigma|$, indicating that the residuals scales with $|1 - \lambda_i/\sigma|$. Again, this factor occurs in the error bounds described in Theorem 3.1.2.

For the ill-conditioned problem, the eigenvalues do not converge as compared to the well-conditioned problem. As shown in Fig 3.4(a), the residual bounds are greatly affected by the conditioning of the problem. This indicates that a non-symmetric factorization of $A - \sigma B$ does not stabilize the Lanczos process.

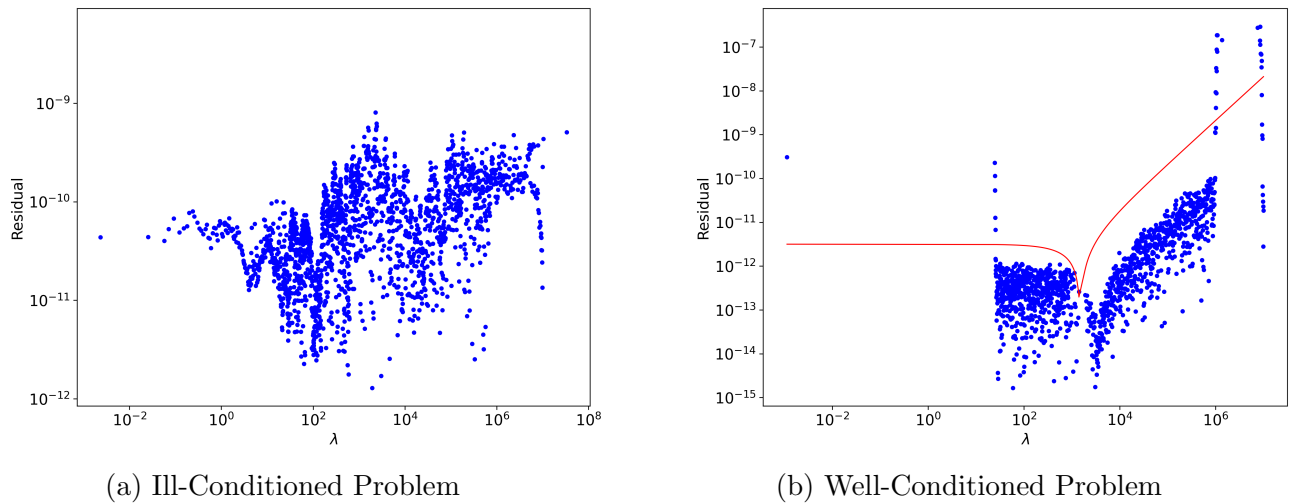


Figure 3.4: Residual plot comparison with a moderate shift (*LU* Decomposition)

CHAPTER 4

CONCLUSION

This thesis has investigated the application and performance of the spectral transformation Lanczos algorithm for solving symmetric definite dense generalized eigenvalue problem. Through the numerical experiments, we validated our results with proven error bounds in direct methods, considered the implication of several factorization techniques, and the impact of certain properties of the matrix on the accuracy of the results. In this concluding chapter, we summarize our key findings, discuss the broader implications of this work, acknowledge limitations, and outline promising directions for future research.

4.1 Summary of Key Findings

The experiments in this thesis have uncovered some interesting results regarding the spectral transformation Lanczos algorithm for generalized eigenvalue problems. First, we have established that a moderate shift σ not too close to a generalized eigenvalue will guarantee small residuals for eigenvalues not much larger than σ , but a large shift will result in having small residuals only for eigenvalues near σ .

Secondly, our analysis of the eigenvalue sensitivity revealed the relationship between the conditioning of the matrices, and the accuracy of computed eigenvalues for various factorizations of the shifted matrix $A - \sigma B$. We observed that for any factorization involving symmetry (eigenvalue decomposition or LDL^T factorization), the ST-Lanczos appears to be stable and the Ritz pairs converged to the order of unit round off u for the n Lanczos steps. The generalized eigenvalues also converged, achieving unit round off for all computed

eigenvalues closer and farther away from the shift. This poses an interesting question: “Can we prove stability for any symmetric decomposition of $A - \sigma B$ ”?

Finally, for the LU decomposition of $A - \sigma B$, we observe that the Lanczos procedure was stable but the behavior is largely dependent on the conditioning of $A - \sigma B$. However, our results indicated that the generalized residuals were insensitive to the conditioning of the problem.

4.2 Importance and Implications

From a theoretical perspective, this work advances our knowledge of spectral transformation, matrix conditioning and eigenvalue sensitivity in the context of generalized eigenvalue problems. Our results showed that the bounds for direct methods, appear to hold true for iterative methods. This work goes a step further at highlighting an interesting property of spectral transformation methods that can determine stability for such methods, both in the direct and iterative context. This contributes to the broader field of numerical linear algebra by providing a more comprehensive framework for analyzing iterative eigenvalue solvers.

By characterizing the relationship between matrix factorizations and algorithm convergence, we have developed a better understanding of how spectral transformations affect the convergence of properties of Krylov subspace methods.

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